MILITARY SPECIFICATION

ADHESIVE, AIRCRAFT, FOR STRUCTURAL REPAIR

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1.1 Scope This specification establishes the requirements for aircraft structural repair adhesives capable of operating within a temperature range of -55°C to +104°C (-67°F to +220°F)

1.2 Classification The adhesives shall be classified as follows.

1.2.1 Types

Type I - Suitable for autoclave curing
Type II - Suitable for vacuum bag curing

1.2.2 Forms

Form F - Film
Form P - Paste (see 6.6)

2 APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and Standards The following specifications and standards form a part of this specification to the extent specified herin Unless otherwise specified, the issue of these documents shall be those listed in the issue of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Systems Engineering and Standardization Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter

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SPECIFICATIONS

FEDERAL

QQ-A-250/4	-	Aluminum Alloy 2024, Plate and Sheet
PPP-B-601	-	Boxes, Wood, Cleated Plywood
PPP-B-636	-	Boxes, shipping, Fiberboard
PPP-C-96	-	Can, Metal, 28 Gage and Lighter

MILITARY

MIL-T-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-A-8243	-	Anti-Icing and Deicing-Defrosting Fluid
MIL-B-22191	-	Barrier Material, Transparent, Flexible Heat Sealable
MIL-L-23699	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-H-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft, NATO Code Number H537
MIL-A-83377	-	Adhesive Bonding (Structural) For Aerospace and Other Systems, Requirements For

STANDARDS

FEDERAL

FED-STD-313	-	Material Safety Data Sheets Preparation and Submission of

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity)

2 2 Other publications The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall those listed in the issue of the DoDISS specified in the solicitation. unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C 297 (Reapproved 1976)	-	Tension Tests of Flat Sandwich Constructions in Flatwise Plane
ASTM D 1781	-	Climbing Drum Peel Test for Adhesives
ASTM D 2912	-	Determining Durability of Adhesive Joints Stressed in Shear by Tension Loading
ASTM D 3166	-	Fatigue Properties of Adhesives in Shear by Tension Loading (Metal-to-Metal)
ASTM D 3167	-	Floating Roller Peel Resistance of Adhesives

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) (Continued)

ASTM D 3528 - Strength Properties of Double Lap Shear Adhesive Joints by Tension Loading
ASTM D 3933 - Preparation of Aluminum Surfaces for Structural Adhesive Bonding (Phosphoric Acid Anodizing)
ASTM D 3983 - Measuring Strength and Shear Modulus of Non-Rigid Adhesives by the Thick Adherend Tensile Lap Specimen

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Institute, 1430 Broadway, New York, NY 10018.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC , AGENT

National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Tariff Order Section, 1616 P Street, N.W , Washington, DC 20036)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents These documents may also be available in or through libraries or other informational services.)

2 3 Order of precedence In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been granted

3 REQUIREMENTS

3 1 Qualification Adhesives furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4 3 and 6 3)

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3 2 Materials

3 2 1 Formulation

3 2.1 1 Chemical structure There shall be no restrictions on the chemical structure of the adhesives except that they shall be thermosetting However, a paste formulation known to meet requirements of this specification is given in 6 6

3 2 1 2 Fillers Fillers may be incorporated into the adhesives, except that asbestos and metal fillers shall not be permitted Any filler used shall be considered part of the adhesive system and shall not have any detrimental effect on the integrity of the bond or the adherends within the operating temperature parameters of the adhesive

3 2 1 3 Scrim Scrim material may be incorporated into the adhesive system. It shall be furnished by the manufacturer and become an integral part of the system The scrim shall not have a detrimental effect on the integrity of the bond or the adherends within the operating limits of the adhesive

3.2 2 Primers Primers may be furnished with the adhesive. The manufacturer shall specify the primer to be used with specific adherends.

3 2 3 Curing agents Curing agents shall be supplied either separately or be incorporated into the adhesive. Separate curing agents shall be of sufficient strength to insure a by weight mixing ratio of not less than 1 part curing agent to 10 parts base resin

3 2 4 Formulation changes An adhesive system shall be approved only for that system on which qualification testing is conducted Any changes to base adhesive, curing agent, and film (including degree of advancement, surface embossment and changing the carrier) shall be cause to remove the approved system from the Qualified Products List

3 3 Forms

3.3 1 Film. Film adhesives shall consist either entirely of adhesive (unsupported) or adhesive with a carrier (supported) The film thickness (in mils) and corresponding weight (in pounds per square foot to the nearest 0 001 pound) shall be stated Film adhesives of a given type and composition manufactured in more than one weight range shall be separately tested however, films of the same type and composition within those weights shall be considered for acceptance

3 3 2 Paste Adhesives in paste form shall mix readily to a smooth solution or suspension of consistency suitable for application and shall be free of lumps The components shall not settle out or separate during a normal working day

3 4 Toxicity The material shall have no adverse effect on the health of personnel when used for its intended purpose in accordance with the directions specified herein The adhesive system shall contain no components which produce noxious vapors in such concentrations as to be an annoyance to

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personnel under conditions of adequate ventilation and while observing Occupational Safety and Health Administration (OSHA guidelines Questions pertaining to the toxic effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.3 2 3 and 6 5)

3 5 Working characteristics

3 5 1 Curing conditions Each type of adhesive shall gel at temperatures less than 90°C (194°F)

3 5 1 1 Type I adhesives Type I adhesives shall be cured for a maximum of 4 hours at a maximum temperature of 149°C (300°F) using an autoclave procedure recommended by the manufacturer that shall not exceed 100 psi

3.5 1.2 Type II adhesives Type II adhesives shall be cured for a maximum of 4 hours at a maximum temperature of 149°C (300°F) and a vacuum of not more than 25 inches of mercury using vacuum bag techniques recommended by the manufacturer.

3.5 2 Advancement Adhesive advancement is permitted The advanced adhesive shall be considered a different system for qualification purposes

3.5.3 Viscosity. The minimum viscosity of the mixed paste adhesive when tested as specified in 4.6.1.1 shall not drop below 100 poises

3 5.4 Bondline thicknesses The adhesives shall be formulated to yield bond line thicknesses of 4 to 10 mils (0 004 to 0.010 inch) when cured as specified in 3.5 1.

3 5 5 Working life The working life of the adhesives, ready and useable for bonding, shall be a minimum of 1 hour

3 6 Mechanical properties Mechanical properties shall be as specified in Table I for normal thickness adherends and Table II for thick adherends

3.7 Storage life. After storage at conditions specified in 4.6 9.1 (Natural aging) or 4 6 9 2 (Accelerated aging), the adhesive shall meet all the requirements of this specification

3 8 Instruction sheet The manufacturer shall provide a dated and titled instruction sheet with each shipment of adhesive (see 5.3) outlining instructions for its use on aluminum, titanium, and composite materials (see 6 1) The instruction sheet shall include the following information

a The general chemical type of the base material used in the adhesive system Identification of the film carrier (supported) Identification of, and instructions for use, of each segment of a multiple-part adhesive system

b Complete procedures for preparation of all adherend surfaces prior to bonding with the adhesive including application of primer, if required

c Complete mixing instructions for the paste adhesive

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d. Typical time, temperature, and pressure for each segment of the complete curing cycle giving maximum and minimum limits for each condition

e Resealing instruction and precautions for packages of adhesive, curing agent and film not completely used in one operation

f Date of manufacture.

3 9 Workmanship Adhesives shall be free of contaminants and shall be prepared in accordance with the best commercial practices for this material Film adhesives shall be free of folds, voids, foreign matter and wrinkles

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.1 1 Responsibility for compliance All items must meet all requirements of sections 3 and 5 The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections The inspection requirements specified herein shall be classified as follows:

a Qualification inspection (see 4 3)
b Quality conformance inspection (see 4 4)

4 3 Qualification inspection The qualification inspection shall consist of all the tests and inspections specified in Table III.

4.3 1 Qualification samples The qualification sample shall consist of 2 quarts of paste or three - 5 square feet sections of film adhesive from different production batches for each type The adhesive shall be packaged as specified in section 5 of this specification Included as part of the qualification sample shall be the required amounts of thinners, curing agents, and primers to yield a complete adhesive system The manufacturer shall use a portion of each adhesive sample to prepare duplicate sets of bonded panels and specimens as specified in 4.3 4 1 through 4 3 4 5 All panels, specimens and

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the remaining adhesive shall be forwarded to the Qualifying Laboratory designated in the authorizing letter The samples shall be plainly and durably marked with the following information:

SAMPLE FOR QUALIFICATION TESTING

ADHESIVE, AIRCRAFT, STRUCTURAL REPAIR

Name and address of manufacturer
Plant address and date of manufacture
Manufacturer's part number and other identification
Batch No of adhesive
Submitted by (name) for qualification in accordance with the requirements Of MIL-A-85705A(AS) under authorization (reference authorizing letter) (See 6.3)

4 3.2 Manufacturer's data to be submitted with sample.

4.3.2.1 Test reports Two copies of the manufacturer's certified test report shall be forwarded along with the qualification samples specified in 4.3 1 The report shall show, by actual test results and specific paragraph references, the adhesive submitted conforms to all the requirements of this specification

4.3.2.2 Instructions for use The manufacturer shall submit 2 copies of the instruction sheet specified in 3 8

4 3.2 3 Toxicological data The manufacturer shall furnish the toxicological data required to evaluate the safety of the material for the proposed use through submission of material safety data sheets prepared in accordance with FED-STD-313

4 3 3 Retention of qualification In order to retain qualification of a product approved for listing on the Qualified Products List (QPL), the manufacturer shall verify by certification to the qualifying activity that the manufacturer's product complies with requirements of this specification The time of periodic verification by certification shall be in two-year intervals from the date of original qualification and shall be initiated by the Government The Government reserves the right to re-examine the qualified product whenever deemed necessary to determine that the product continues to meet any or all of the specification requirements.

4 3 4 Preparation of test specimens All aluminum test panels shall be surface treated in accordance with ASTM D 3933 prior to the bonding operation All edges of the metal panels and specimens which will be within (or which will bond) the joints shall be machined true (without burrs or bevels and at right angles to faces) and smooth (RHR 140 maximum, in accordance with ANSI B46 1) before the panels are surface treated and bonded Panels shall not be cut into specimens until at least 24 hours after bonding Aluminum-alloy panels may be sawed with a bandsaw, the setting and spacing of teeth and operational speed adjusted to hold frictional heating of the bond to a minimum All panels and specimens shall be so marked that each specimen can be identified at any time with the particular panel from which it was cut and with the particular batch of adhesive with which it was bonded so that any failure may be properly traced to either the adhesive or the mechanics or equipment of bonding

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4.3.4.1 Double lap-shear specimens. Type B specimens in ASTM D 3528 shall be prepared from aluminum alloy 2024-T3 conforming to QQ-A-250/4. The panel dimensions shall be such that 6 test specimens shall be obtained from each panel The specimens shall have a 0.50 inch (1.27 cm) overlap and an overall length of 7.5 inches (19.1cm). Double lap shear specimens are required for tests 1 through 4 and 13 through 17 Fifty-four specimens are required (9 panels).

4.3 4.2 Peel resistance specimens. Peel resistance specimens for tests 5 through 8 of Table I shall be prepared in accordance with ASTM D 3167. The panel dimensions shall be such that six- 0.5 inch (1.27 cm) test specimens shall be tested from each panel, after discarding the ends of the panel Twenty-Four test specimens (4 panels) are required

4.3 4.3 Flatwise tensile specimens. Butt joint specimens for tests 9 through 12 shall be prepared by bonding the adhesive directly to aluminum alloy 2024-T3 blocks conforming to QQ-A-250/4 The block specimen shall be configured as shown in ASTM C 297, except that no core material shall be used Each specimen (2 blocks plus adhesive) shall be assembled so that the holes in the loading blocks are 90 degrees to each other and the sides of the block are aligned. A total of twenty-four specimens are required.

4 3 4 4 Thick adherend lap shear specimens. Thick adherend specimens shall be prepared as shown in Figure 1. Forty-two specimens are required for tests 18 through 22 in Table II.

4 3 4.5 Climbing drum peel specimens. Climbing drum peel specimens for tests 23, 24, and 25 shall be prepared in accordance with ASTM D 1781. A total of 18 specimens are required for these tests

4 4 Quality conformance inspection

4 4 1 Lot formation. A lot shall consist of all the adhesive film of one type or 500 pounds (227 Kg) or fraction thereof of adhesive paste manufactured at one time from the same batch of adhesive (see 6 4) and submitted for inspection at the same time and place.

4 4.2 Sampling

4 4.2 1 For visual inspection. All films shall be visually inspected as specified in 4.4.3 1

4.4 2 2 For physical property tests A sample of not less than 2 pints of paste or an area of film sufficient to accomplish all testing shall be taken at random from each lot offered for inspection and tested to the requirements specified in 4 4 3 2

4 4 2 3 Packaging.

4 4 2 3 1 Unit packaging A random sample of unit containers shall be selected from each lot in accordance with Inspection Level I of MIL-STD-105 and acceptable Quality Level of 2 5 percent defective for an inspection as specified in 4 4.3 3 The sample unit shall be one unit package The lot size shall be the number of unit packages

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4 4.2.3 2 Shipping containers Shipping containers, just prior to closure, shall be randomly selected from each lot in accordance with Inspection Level I and Acceptable Quality Level of 4 0 percent defective of MIL-STD-105. The sample unit shall be one shipping container. The lot size shall be the number of shipping containers. Inspection shall be as specified in 4 4 3.3

4.4 3 Quality conformance tests and inspection

4 4.3 1 Visual inspection. All film in the lot shall be visually inspected to the requirements specified in 3.9.

4 4 3.2 Physical property tests. The sample selected in 4 4.2 2 shall be tested for thick adherend shear stress at 24°C (75°F) (Test # 18) and at 104°C (220°F) after a 10 minute soak, (Test # 19). Two test panels shall be prepared (4.3 4 4) for each test Elastic modulus, shear strength and shear strain shall be reported All available specimens shall be tested Failure of any specimen to meet or exceed the requirement specified in Table II shall be cause to reject the lot represented by the specimen.

4 4.3 3 Packaging inspecton Samples selected in accordance with 4.4.2.3.1 and 4.4 2 3 2 shall be examined to the applicable requirements in Table IV and all other applicable requirements to determine conformance to section 5 of this specification. In addition, shipping containers fully prepared for delivery shall be inspected for closure defects

4 5 Test conditions

4 5 1 Normal temperature. Normal temperature tests, when specified herein, shall be conducted at 24° ± 3°C (75° ± 5°F). Testing shall be initiated 10 +2, -0 minutes after the specimens have reached equilibrium at this temperature The required time for the specimen to reach normal temperature equilibrium following conditioning shall not be less than 10 minutes nor more than 30 minutes

4.5 2 Elevated temperature The final testing temperature (104°C) of all specimens for elevated temperature tests shall be those of the surface of the metal as close to the bonded area as possible. The temperature shall be determined with a thermocouple, firmly attached at the bond line of the specimens, in order to insure accuracy of the testing temperature and reproducibility of data. Elevated temperature tests shall be conducted 10 +2, -0 minutes after the specimens in the testing chamber have reached temperature equilibrium at the specified temperature The time required for the specimens to reach equilibrium shall be 45 ± 10 minutes

4 5 3 Low temperature Low temperature tests shall be conducted at -55° ± 3°C (-67° ± 5°F). Testing shall be initiated 10 +2, -0 minutes after the specimens have reached equilibrium at that temperature The method for determining the temperature of the specimens and the time required for the specimens to reach temperature equilibrium shall be in accordance with 4.5.2

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4 5 4 High humidity condition. Resistance to humidity exposure shall be determined after 14-days ± 2-hours exposure of the specimens in a humidity chamber maintained at 77° ± 3°C (170° ± 5°F) and 95 to 100 percent relative humidity The temperature in the chamber shall be recorded at least twice each working day. Condensate from one panel shall not drip upon any other panel Distilled water, or water containing not more than 200 parts per million of total solids shall be used to maintain the humidity. The test shall be continuous for the duration of the 14 day period Continuous operation means that the chamber shall be closed and the humidity maintained continuously, except for the short daily interruptions necessary to inspect rearrange, remove test panels,or to check or replenish the water in the reservoir. The test specimens shall be individually suspended vertically in the atmosphere above the water and shall not contact the sides of the container or other test specimens

4 5 5 Fluid immersion conditions Test specimens shall be immersed in the fluids specified in Table V. Immersion conditions shall be 49° ± 3°C (120° ± 5°F) for 7 days ± 2 hours All specimens shall be arranged in the fluid container to insure full contact with the fluid while not contacting the other specimens. Testing of the specimens shall be conducted at 104° ± 3°C (220° ± 5°)

4 6 Test methods

4.6 1 Working characteristics.

4 6 1 1 Viscosity and working life (Form P only) Viscosity of the mixed adhesive shall not drop below 100 poises when heated at a rate of 2°C(4°F) A Brookfield viscometer Model RVT (or equivalent instrument) operated at 1 rpm with a number 7 spindle shall be used

4.6 1.2 Cure All panels shall be cured as specified in 3 5 1 and the manufacturer's instruction sheet.

4 6 2 Double Lap tensile shear

4.6.2 1 Normal temperature testing (test 1) Double lap tensile shear shall be determined in accordance with ASTM D 3528 Six specimens shall be tested at 24° ± 3°C (75° ± 5°F) after conditioning specified in 4.5 1. All values shall be reported including the mean and standard deviation(s)

4.6.2.2 Elevated temperature (dry) (test 2). Elevated temperature testing and reporting of results shall be as specified in 4 6.2.1 except that testing shall be conducted at 104° ± 3°C (220° ± 5°F) after conditioning specified in 4 5.2

4.6 2 3 Elevated temperature (wet) (test 3) Double lap shear specimens shall be exposed to the wet conditions of 4 5 4, and tested wet, at 104° ± 3°C (220° ± 5°F). Test procedure and reporting of results shall be as specified in 4.6.2.1

4 6 2.4 Low temperature test (test 4) Low temperature test procedure of double lap shear specimens and reporting of results shall be in accordance with 4 6.2 1 except that testing shall be at -55° ± 3°C (-67° ± 5°F) after conditioning specified in 4 5.3

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4.6.3 Peel resistance

4.6 3 1 Normal temperature testing (test 5) Peel resistance testing of the adhesive shall be in accordance with ASTM D 3167. Six specimens shall be tested at 24° ± 3°C (75° ± 5°F) after conditioning specified in 4.5 1. All values shall be reported along with the mean and standard deviation(s).

4 6 3.2 Elevated temperature (dry) (test 6) Elevated temperature testing and reporting of results shall be as specified in 4.6.3.1 except that testing shall be at 104° ± 3°C (220° ± 5°F) after conditioning of 4.5.2.

4 6 3.3 Elevated temperature (wet) (test 7). Peel resistance specimens shall be exposed to the humidity condition of 4 5 4, then tested wet, at 104° ± 3°C (220° ± 5°F) in accordance with 4.6.3.1. Reporting of results shall be as specified in 4 6.3.1.

4 6 3 4 Low temperature test (test 8). Low temperature testing and reporting of results shall be in accordance with 4 6 3 1 except that testing shall be at -55° ± 3°C (-67° ± 5°F) after conditioning as specified in 4.5 3

4 6 4 Flatwise tensile testing

4 6 4 1 Normal Temperature (test 9). Flatwise tensile determinations shall be performed in accordance with ASTM C 297 and the following: Test specimens shall be in accordance with 4.3.4.3. Tensile load shall be applied to the fixture at a constant rate of movement of the moveable head of the testing machine of not less than 0 015 inch (.38 mm) and not more than 0.020 inch (51 mm) per minute until failure Six specimens shall be tested at 24° ± 3°C (75° ± 5°F) after conditioning in 4 5 1 All results shall be reported along with the mean and standard deviation(s).

4.6 4.2 Elevated temperature (dry) (test 10). Elevated temperature testing and reporting of results shall be in accordance with 4.6 4 1 except that testing shall be at 104° ± 3°C (220° ± 5°F) after conditioning of 4.5 2

4 6 4 3 Elevated temperature (wet) (test 11) Six specimens shall be exposed to the humidity conditions of 4.5 4, then tested wet, at 104° ± 1°C (220° ± 2°F) as specified in 4.6.4.1.

4.6 4 4 Low temperature test (test 12). Low temperature testing and reporting of results shall be in accordance with 4.6.4.1 except that testing shall be -55° ± 3°C (-67° ± 5°F) after the conditioning specified in 4 5.3

4 6 5 Static Long term stress durability (test 13) Six specimens prepared in 4 3 4 1 shall be loaded to 1500 psi (10 4MPa) and exposed to 60° ± 1°C (140° ± 2° F) and 95-100% relative humidity Exposure time shall be 90 days There shall be no failure Creep measurements shall be made upon completion of the exposure period with specimens still under load at the exposure temperature Deformation shall not exceed 0.0075 inch (0.019cm) on bonded joints of all specimens

4 6 6 Cyclic stress durability (test 14) Six specimens prepared in accordance with 4 3 4 1 shall be cyclic loaded at 60° ± 2°C (140° ± 5°F) and 95 to 100 percent relative humidity as follows. 50 minutes at 1500 psi (10 4MPa), 10 minutes at rest Specimens shall complete 300 cycles without failure.

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4.6 7 Fatigue Strength

4 6 7.1 Normal temperature (test 15). Six specimens prepared as specified in 4.3.4.1 shall be subjected to fatigue testing at 24° ± 3°C (75° ± 5°F) in accordance with ASTM D 3166 at 180 cycles/minute, with R = 0.1 at 2000psi (13 8MPa). There shall be no failures at 10^6 cycles.

4 6 7.2 Elevated temperature (test 16) Six specimens prepared as specified in 4.3.4.1 shall be subjected to fatigue testing at 104° ± 3°C (220° ± 5°F) as specified in ASTM D 3166 at 180 cycles/minute, with R = 0 1 at 1200psi (8 3MPa). There shall be no failure at 10^6cycles.

4.6 7.3 Low temperature (test 17) Six specimens prepared as specified in 4 3.4 1 shall be subjected to fatigue testing at -55° ± 3°C (-67° ± 5°F) in accordance with ASTM D 3166 at 180 cycles/minute, with R = 0 1 at 2000(psi) (13.8MPa). There shall be no failures at 10^6 cycles

4 6 8 Thick adherend lap shear test

4 6 8 1 Normal temperature (test 18) Six specimens prepared as specified in 4 3 4.4 shall be tested for elastic modulus, shear strength and shear strain after conditioning in accordance with 4.5 1. Test procedure shall be in accordance with ASTM D 3983 All results shall be reported along with the mean and standard deviation(s).

4 6 8 2 Elevated temperature (dry) (test 19). Elevated temperature shear property testing and reporting of results shall be as specified in 4.6.8.1 except that testing shall be at 104° ± 3°C (220° ± 5°F) after conditioning in 4.5.2 at 104° ± 3°C

4 6 8 3 Low temperature testing (test 20). Low temperature thick adherend shear testing and reporting of results shall be as specified in 4.6.8 1 except that testing shall be -55° ± 3°C (-67° ± 5°F) after conditioning as specified in 4 5 3

4.6 8.4 Elevated temperature testing after humidity exposure (test 21). Thick adherend specimens shall be subjected to the humidity exposure of 4.5.4 followed by testing, wet at 104° ± 3°C (220° ± 5°F). Testing and reporting of results shall be in accordance with 4.6.8.1.

4.6.8 5 Elevated temperature testing after fluid immersion (test 22). Six thick adherend specimens (4 3 4 4) shall be immersed in each of the fluids specified in 4.5.5 Exposure time and temperature shall be in accordance with 4 5 5 Tests shall be conducted at 104° ± 3°C (220° ± 5°F) within six hours after removal from the fluid Test procedure and reporting of results shall be as specified in 4.6 8 1

4 6 9 Storage stability

4.6 9 1 Natural aging Film adhesives (Form F) shall be stable for at least 12 months at temperatures not above -17 8°C (0°F) from date of shipment in the original unopened containers Paste adhesives (Form P) shall be stable

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for at least 12 months at temperatures between 27° and 32°C (80° and 90°F) from date of shipment All adhesives systems shall be capable of meeting all the requirements of this specification after storage. The above storage conditions are for qualification testing purposes.

4.6 9 2 Accelerated storage The paste adhesive shall be stored at 38°±1°C (100°±2°F) for a period of six months Conformance to 3 7 shall be noted The above storage conditions are for qualification testing purposes

4 6.10 Climbing drum peel test

4 6.10 1 Normal temperature testing(test 23) Peel testing shall be in accordance with ASTM D 1781 Five specimens shall be tested at 24°±3°C (75°±5°F). All test results shall be reported along with the mean and standard deviation.

4.6.10.2 Elevated temperature(dry)(Test 24) Elevated temperature testing and reporting of results shall be as specified in 4 6 10 1, except that testing shall be at 104°±3°C(220°±5°F) after conditioning as specified in 4.5 2.

4.6.10.3 Low temperature testing (Test 25) Low temperature testing and reporting of results shall be as specified in 4.6.10 1 except that testing shall be at -54°±5°C(-65°±5°F)

5 PACKAGING

5 1 Preservation-packaging Preservation - packaging shall be level A or Commercial, as specified (see 6.2)

5 1 1 Level A.

5 1.1 1 Paste adhesives.

5.1.1.1.1 Unit packaging. Paste adhesives shall be packaged in two part, flexible, divider packages which separate the components until mixing. Each package shall contain between 50 to 70 grams of A side(see 6 6) and the appropriate amount of curing agent. An equivalent method of packaging (such as a dispenser type) may be used when specified by the acquisition activity (See 6.2.1). The packaging material shall not react with nor be degrading to the adhesive or curing agent.

5.1 1 1 2 Intermediate packaging Paste adhesives, packaged as specified in 5 1 1 1 1, shall be intermediate packaged in accordance with the appendix of PPP-C-96, with containers conforming to PPP-B-636, grade V3c and shall be within the weight limitations specified for the container Closure and waterproof sealing of the boxes shall be in accordance with the appendix to PPP-B-636

5.1 1.2 Film adhesive

5 1 1.2 1 Unit packaging Film adhesive, in roll form as specified (see 6.2), shall be wrapped in a suitable release paper, enclosed in vapor barrier

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material in accordance with MIL-B-22191, Type I or MIL-B-131, Class 1 and individually packaged in boxes conforming to PPP-B-636, grade W6c or grade W6s Each box shall be waterproofed as specified for the weatherproofing of slotted boxes in accordance with the appendix of PPP-B-636 The film adhesive is to be supplied in rolls of specified width (inches), length (feet), thickness (mils), and weight (pounds per square foot)

5 1.1 2.2 Intermediate packaging Twelve rolls of film adhesive, packaged as specified in 5 1 1 2 1, shall be intermediate packaged in boxes conforming to PPP-B-636, V3c

5 1.2 Commercial The adhesive, paste or film, shall be packaged in accordance with normal commercial practice When two adhesive components are necessary as a unit, the components shall be packaged as a kit in a single container (plastic, separated bag) The complete package shall protect the adhesive against damage during shipment, handling and storage

5.2 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.2 1 Level A. Adhesives, intermediate packaged as specified in 5.1.1.1.2 or 5.1.1 2 2, shall be packed in boxes conforming to PPP-B-601, overseas type, style optional in quantities not to exceed the weight limitation of the container. Box closure and strapping shall be in accordance with the appendix to the box specification Strapping shall be zinc coated

5 2.2 Level B. Adhesive, intermediate packaged as specified in 5.1 1 1 2 or 5 1 1 2 2, shall be shipped directly in the intermediate container, no additional overpack is necessary. Boxes shall be strapped in accordance with requirements in the appendix to PPP-B-636

5.2.3 Commercial Adhesive, packaged as specified in 5 1, shall be packed to assure carrier acceptable and safe delivery to destination at lowest rating in conformance to requirements of Uniform Freight Classification or National Motor Freight Classification

5.3 Marking.

5.3 1 Film adhesive Each film adhesive vapor barrier containers shall be marked to include the following "Upon removal from freezer, allow adhesive to come to room temperature and all condensation on the vapor barrier material to evaporate before removing adhesive from the vapor barrier material

5 3 2 Interior and exterior containers In addition to markings required by the contract or order, the interior and exterior shipping containers shall be marked in accordance with MIL-STD-129 and shall include the following

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ADHESIVE, AIRCRAFT, FOR STRUCTURAL REPAIR
MIL-A-85705A(AS)
Type and form
Date of manufacture
Manufacturer's name and address
Manufacturer's product designation
Manufacturer's batch/lot identification
Contract number
Date of shipment
Expiration date
Manufacturer's recommended storage temperature range
Quantity contained
Film weight (when applicable)
For Films-store @ 0°F

6 NOTES

6 1 Intended use. Adhesives conforming to the specification are intended for use in repair of aircraft structures and subassemblies Included are metal-to-metal, composite-to-composite, and composite-to-metal combinations

6.2 Ordering data

6 2 1 Acquisition requirements Acquisition documents should specify the following

a. Title, number and date of this specification
b Type and form
c Amounts and unit quantities required
d. Length, width, thickness and weight (psf) of film adhesive roll
e Level of packaging and packing required (see 5 1 and 5.2)
f Paste packaging, if other than specified in 5 1 1 1 1

6.3 Qualification With respect to products requiring qualification, awards will be made only for products which are at the time set for opening bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification The activity responsible for the Qualified Products List is the Naval Air Systems Command (AIR-5304C2), Washington, DC 20361; however, information pertaining to qualification of products and letter of authorization for submittal of samples may be obtained from the Commanding Officer Naval Air Engineering Center, Lakehurst, NJ 08733-5100 Atterntion Systems Engineering and Standardization Department (Code 9321)

6 4 Batch A batch shall consist of a homogeneous unit of adhesive manufactured at one time or representing a blend of several manufactured units of adhesive of the identical formulation which is used to prepare a lot of adhesive.

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6 5 Toxicity. Questions pertaining to the requirement of 3 4, should be referred by the acquiring activity, to the Navy Environmental Health Center, Naval Station, Norfolk, VA 23511, who will act as an advisor to the acquiring activity.

6 6 Adhesive paste formulation. The paste adhesive used to generate the requirements of the specification was formulated from a diglycidyl ether bisphenol A type epoxy(DGEBA) modified with a carboxy terminated acrylonitrile/butadiene rubber(CTBN) as follows:

A side

240 0 grams -DGEBA
54 0 grams - CTBN
5.0 grams - cycloaliphatic diamine (Amicure PACM, Air Products Co)
0 6 grams - triphenylphosphene

The above shall be reacted as follows

Heat the DGEBA and CTBN, while stirring, to 160°C(320°F). Add the triphenylphosphene, heat to 170°C(338°F) and hold for one hour. Add the cycloaliphatic diamine and continue heating at 170°C(338°) for an additional hour Cool to room temperature

B side

340 0 grams - cycloaliphatic diamine (Amicure PACM, Air Products Co)
100 0 grams - DGEBA

Mix the diamine and the DGEBA at room temperature Heat to 120°C(248°F) and hold for one hour Cool to room temperature

10 parts by weight of A side shall be mixed with 3.8 parts by weight of B side to form the compound used for tests and evaluation.

6.7 Subject term (Key word) listing

Adhesive
Aircraft
Qualification
Structural repair

6 7 Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

Preparing Activity
Navy-AS
(Project No 8040-N130)

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Table I Mechanical Properties - Normal Thickness Adherends

Properties	Test para	Requirements 1/	Test Specimens	Test No
Double lap shear psi (MPa), min	4 6 2 1	5000 (34 5)	6	1
	4.6.2 2	3000 (20 7)	6	2
	4.6.2 3	2300 (15 9)	6	3
	4.6.2 4	5000 (34 5)	6	4
Peel resistance, lbf/in (N/m), min	4 6.3 1	30 (5250)	6	5
	4 6.3 2	25 (4375)	6	6
	4.6.3.3	25 (4375)	6	7
	4 6 3 4	25 (4375)	6	8
Flatwise tensile psi (MPa), min	4 6 4 1	1000 (6.89)	6	9
	4 6.4 2	650 (4.48)	6	10
	4 6.4 3	650 (4.48)	6	11
	4 6.4 4	1000 (6.89)	6	12
Static long term stress durability	4.6.5	No failure, after 90 days	6	13
Creep, inch (cm) max		0.0075 (1.90cm)		
Cyclic stress	4.6.6	No failure, 300 cycles	6	14
Fatigue strength	4.6.7.1	No failure 10^6 cycles	6	15
	4 6.7 2	No failure 10^6 cycles	6	16
	4 6 7 3	No failure 10^6 cycles	6	17
Climbing drum peel inlb/in( ), min	4.6.10			
	4.6.10.1	14	6	23
	4.6.10.2	14	6	24
	4 6 10 3	16	6	25

1/ All failures shall be examined to identify cause of failure (cohesive/adhesive, lack of adhesive contact)

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Table II Mechanical Properties - Thick Adherends

Properties	Test para	Requirements 1/	Test No
Single lap shear			
@ normal temp	4 6 8 1		18
Initial elastic modulus, psi(MPa)		70,000 to 140,000 (483 to 965)	
Ultimate shear stress, psi (MPa), min		4,800 (33 1)	
Ultimate shear strain, in/in(mm/mm), min		.3	
@ 104° ± 3°C(220° ± 5°F)	4 6 8 2		19
Initial elastic modulus, psi(MPa)		30,000 to 100,000 (207 to 690)	
Ultimate shear stress, psi (MPa), min		3,000 (20.7)	
Ultimate shear strain, in/in(mm/mm), min		.8	
@ -55° ± 3°C(-67° ± 5°F)	4.6 8 3		20
Initial elastic modulus, psi(MPa)		70,000 to 140,000 (483 to 965)	
Ultimate shear stress, psi (MPa), min		4,800 (33.1)	
Ultimate shear strain, in/in(mm/mm), min		.2	
@ 104° ± 3°C(220° ± 5°F) after 4.5 4	4 6 8 4		21
Initial elastic modulus, psi(MPa)		30,000 to 100,000 (207 to 690)	
Ultimate shear stress, psi (MPa), min		2700 (18 6)	
Ultimate shear strain, in/in(mm/mm), min		6	
@ 104° ± 3°C(220° ± 5°F) after 4.5 5	4 6 8 5		22
Initial elastic modulus, psi(MPa)		30,000 to 140,000 (207 to 965)	
Ultimate shear stress, psi (MPa), min		2700 (18 6)	
Ultimate shear strain, in/in(mm/mm), min		6	

1/ All failures shall be examined to identify cause of failure (cohesive/adhesive, lack of adhesive contact)

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Table III Qualification Inspection

Curing	3.5.1	4.6.1
Viscosity	3.5.3	4.6.1
Working time	3.5.5	4.6.1
Double lap tensile shear	Table I	4.6.2
Peel resistance	Table I	4.6.3
Flatwise tensile strength	Table I	4.6.4
Long term stress durability	Table I	4.6.5
Cyclic stress durability	Table I	4.6.6
Fatigue strength	Table I	4.6.7
Lap tensile shear thick adherend		
Initial elastic modulus	Table II	4.6.8
Ultimate shear strength	Table II	
Ultimate shear strain	Table II	
Storage stability	3.7	4.6.9
Climbing drum peel	Table I	4.6.10

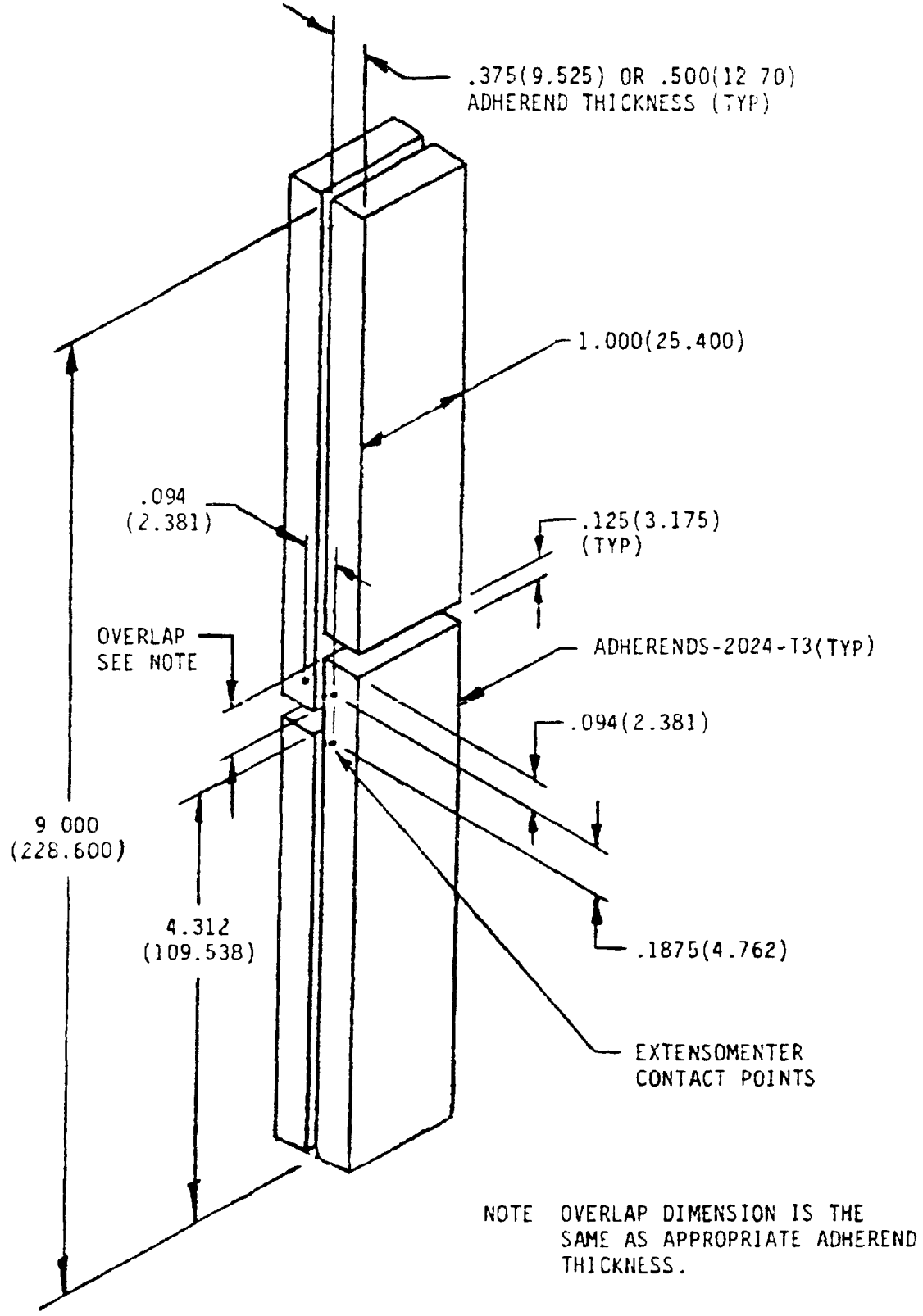
TABLE IV Packaging examination

Packaging	Material or construction not as specified Components damaged, missing, or film not protected Unit package closure incomplete or damaged Not level required by contract or purchase order
Packing	Not level required by contract or purchase order Any nonconforming component, incomplete closures Bulged or damaged shipping containers
Count	Less than specified or indicated quantity per shipping container.
Markings	Unit package film barrier material and packing-omitted, illegible, incorrect, incomplete or not in accordance with contract requirements Instruction sheet missing

TABLE V Immersion fluids

Fluid	Specification
Hydraulic fluid	MIL-H-83282
JP-4 fuel	MIL-T-5624
Lubricating oil	MIL-L-23699
Anti-icing fluid	MIL-A-8243

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NOTE OVERLAP DIMENSION IS THE SAME AS APPROPRIATE ADHEREND THICKNESS.

DIMENSIONS IN INCHES(mm)

Figure 1. Thick adherend lap shear specimen

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-83795A	ADHESIVE, AIRCRAFT, FOR STRUCTURAL REPAIR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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